

Unit 12: Permutation and Combination, Probability

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Objectives

The key concepts explained in this unit, including area a learner is expected to learn and master after going through the unit are: -

- Understand what are the basics of counting.
- Understand the use of product rule in the basics of counting.
- Understand the use of Sum rule in the basics of counting.
- Understand the use of the Subtraction rule in the basics of counting.
- Understand the use of tree diagram in the basics of counting.
- Understand the Division rule in the basics of counting.
- Understand how to find the total number of functions in the basics of counting.

Introduction

Suppose that a password on a computer system consists of six, seven, or eight characters. Each of these characters must be a digit or a letter of the alphabet. Each password must contain at least 1 one digit. How many such passwords are there? The techniques needed to answer this question and a wide variety of other counting problems will be introduced in this section.

Counting problems arise throughout mathematics and computer science. For example, we must count the successful outcomes of experiments and all the possible outcomes of these experiments to determine the probabilities of discrete events. We need to count the number of operations used by an algorithm to study its time complexity.

We will introduce the basic techniques of counting in this section. These methods serve as the foundation for almost all counting techniques. We first present two basic counting principles, the

product rule and the sum rule. Then we will show how they can be used to solve many different counting problems. The product rule applies when a procedure is made up of separate tasks.

12.1 The Product Rule

Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure. Examples 1–10 show how the product rule is used.



Example 1

A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution: The procedure of assigning offices to these two employees consists of assigning an office to Sanchez, which can be done in 12 ways, then assigning an office to Patel different from the officer assigned to Sanchez, which can be done in 11 ways. By the product rule, there are $12 \cdot 11 = 132$ ways to assign offices to these two employees.



Example 2

The chairs of an auditorium are to be labelled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labelled differently?

Solution: The procedure of labelling a chair consists of two tasks, namely, assigning to the seat one of the 26 uppercase English letters, and then assigning to it one of the 100 possible integers.

The product rule shows that there are $26 \cdot 100 = 2600$ different ways that a chair can be labelled. Therefore, the largest number of chairs that can be labelled differently is 2600.



Example 3

There are 32 microcomputers in the computer centre. Each microcomputer has 24 ports. How many different ports to a microcomputer in the centre are there?

Solution: The procedure of choosing a port consists of two tasks, first picking a microcomputer and then picking a port on this microcomputer. Because there are 32 ways to choose the micro-computer and 24 ways to choose the port no matter which microcomputer has been selected, the product rule shows that there are $32 \cdot 24 = 768$ ports.

An extended version of the product rule is often useful. Suppose that a procedure is carried out by performing the tasks $T_1, T_2, \dots, T_m$ in sequence. If each task T_i , $i = 1, 2, \dots, m$, can be done in n_i ways, regardless of how the previous tasks were done, then there are $n_1 \cdot n_2 \cdot \dots \cdot n_m$ ways to carry out the procedure. This version of the product rule can be proved by mathematical induction from the product rule for two tasks.



Example 4

How many different bit strings of length seven are there?

Solution:

Each of the seven bits can be chosen in two ways because each bit is either 0 or 1. Therefore, the product rule shows there are a total of $2^7 = 128$ different bit strings of length seven.



Example 5

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How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene)?

----	-----
26 choices	10 choices
for each	for each
letter	digit

Solution:

There are 26 choices for each of the three uppercase English letters and ten choices for each of the three digits. Hence, by the product rule there are a total of $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$ possible license plates.



Example 6 Counting Functions How many functions are there from a set with m elements to a set with

n elements?

Solution:

A function corresponds to a choice of one of the n elements in the codomain for each of the elements in the domain. Hence, by the product rule, there are $n \cdot n \cdot \cdots \cdot n = n^m$ functions from a set with m elements to one with n elements. For example, there are $5^3 = 125$ different functions from a set with three elements to a set with five elements.



Example 7

Counting One-to-One Functions How many one-to-one functions are there from a set with m elements to one with n elements?

Solution:

First, note that when $m > n$ there are no one-to-one functions from a set with m elements to a set with n elements.

Now let $m \leq n$. Suppose the elements in the domain are $a_1, a_2, \dots, a_m$. There are n ways to choose the value of the function at a_1 . Because the function is one-to-one, the value of the function at a_2 can be picked in $n - 1$ way (because the value used for a_1 cannot be used again).

In general, the value of the function at a_k can be chosen in $n - k + 1$ ways. By the product rule, there are $n(n - 1)(n - 2) \cdots (n - m + 1)$ one-to-one functions from a set with m elements to one with n elements. For example, there are $5 \cdot 4 \cdot 3 = 60$ one-to-one functions from a set with three elements to a set with five elements.

12.2 More Complex Counting Problems

Many counting problems cannot be solved using just the sum rule or just the product rule. However, many complicated counting problems can be solved using both of these rules in combination. We begin by counting the number of variable names in the programming language BASIC. (In the exercises, we consider the number of variable names in JAVA.) Then we will count the number of valid passwords subject to a particular set of restrictions.



Example 15

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In a version of the computer language BASIC, the name of a variable is a string of one or two alphanumeric characters, where uppercase and lowercase letters are not distinguished. (An alphanumeric character is either one of the 26 English letters or one of the 10 digits.) Moreover, a variable name must begin with a letter and must be different from the five strings of two characters that are reserved for programming use. How many different variable names are there in this version of BASIC?

Solution:

Let V equal the number of different variable names in this version of BASIC. Let v_1 be the number of these that are one character long and v_2 be the number of these that are two characters long. Then by the sum rule, $V = v_1 + v_2$. Note that $v_1 = 26$, because a one-character variable name must be a letter. Furthermore, by the product rule, there are $26 \cdot 36$ strings of length two that begin with a letter and end with an alphanumeric character. However, five of these are excluded, so $v_2 = 26 \cdot 36 - 5 = 931$. Hence, there are $V = v_1 + v_2 = 26 + 931 = 957$ different names for variables in this version of BASIC.

**Example 16**

Each user on a computer system has a password, which is six to eight characters long, where

each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution:

Let P be the total number of possible passwords, and let p_6 , p_7 , and p_8 , denote the number of possible passwords of length 6, 7, and 8, respectively. By the sum rule,

$P = p_6 + p_7 + p_8$. We will now find $p_6 + p_7$ and p_8 . Finding p_6 directly is difficult. To find p_6 it is easier to find the number of strings of uppercase letters and digits that are six characters long, including those with no digits, and subtract from this the number of strings with no digits. By the product rule, the number of strings of six characters is 366, and the number of strings with no digits is 266. Hence,

$p_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560$. Similarly, we have $p_7 = 36^7 - 26^7 = 78,364,164,096 - 8,031,810,176 = 70,332,353,920$ and $p_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880$.

Consequently, $P = p_6 + p_7 + p_8 = 2,684,483,063,360$.

**Example 17**

Counting Internet Addresses On the Internet, which is made up of interconnected physical networks of computers, each computer (or more precisely, each network connection of a computer) is assigned an Internet address. In Version 4 of the Internet Protocol (IPv4), now in use,

Bit number	0	1	2	3	4	8	16	24	31
Class A	0	netid				hostid			
Class B	1	0	netid				hostid		
Class C	1	1	0	netid				hostid	
Class D	1	1	1	0	Multicast address				
Class E	1	1	1	1	0	address			

. an address is a string of 32 bits. It begins with a network number (netid). The netid is followed by a host number (hostid), which identifies a computer as a member of a particular network. Three forms of addresses are used, with different numbers of bits used for netids and hostids. Class A addresses, used

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for the largest networks, consist of 0, followed by a 7-bit netid and a 24-bit hostid. Class B addresses, used for medium-sized networks, consist of 10, followed by a 14-bit netid and a 16-bit hostid. Class C addresses, used for the smallest networks, consist of 110, followed by a 21-bit netid and an 8-bit hostid. There are several restrictions on addresses because of special uses: 1111111 is not available as the netid of a Class A network, and the hostids consisting of all 0s and all 1s are not available for use in any network. A computer on the Internet has either a Class A, a Class B, or a Class C address. (Besides Class A, B, and C addresses, there are also Class D addresses, reserved for use in multicasting when multiple computers are addressed at a single time, consisting of 1110 followed by 28 bits, and Class E addresses, reserved for future use, consisting of 11110 followed by 27 bits. Neither Class D nor Class E addresses are assigned as the IPv4 address of a computer on the Internet.)

Figure 1 illustrates the IPv4 addressing. (Limitations on the number of Class A and Class B netids have made IPv4 addressing inadequate; IPv6, a new version of IP, uses 128-bit addresses to solve this problem.)

How many different IPv4 addresses are available for computers on the Internet?

Solution:

Let x be the number of available addresses for computers on the Internet, and let x_A, x_B, x_C denote the number of Class A, Class B, and Class C addresses available, respectively.

By the sum rule, $x = x_A + x_B + x_C$

To find x_A , note that there are $2^7 - 1 = 127$ Class A netids, recalling that the netid 1111111 is unavailable. For each netid, there are $2^{24} - 2 = 16,777,214$ hostids, recalling that the hostids consisting of all 0s and all 1s are unavailable. Consequently, $x_A = 127 \cdot 16,777,214 = 2,130,706,178$.

To find x_B and x_C , note that there are $2^{14} - 2 = 16,384$ Class B netids and $2^{21} - 2 = 2,097,152$ Class C netids. For each Class B netid, there are $2^{16} - 2 = 65,534$ hostids, and for each Class C netid, there are $2^8 - 2 = 254$ hostids, recalling that in each network the hostids consisting of all 0s and all 1s are unavailable. Consequently, $x_B = 1,073,709,056$ and $x_C = 532,676,608$. We conclude that the total number of IPv4 addresses available is $x = x_A + x_B + x_C = 2,130,706,178 + 1,073,709,056 + 532,676,608 = 3,737,091,842$.



Example 18

How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution:

We can construct a bit string of length eight that either starts with a 1 bit or ends with the two bits 00, by constructing a bit string of length eight beginning with a 1 bit or by constructing a bit string of length eight that ends with the two bits 00. We can construct a bit string of length eight that begins with a 1 in $2^7 = 128$ ways. This follows by the product rule because the first bit can be chosen in only one way and each of the other seven bits can be chosen in two ways. Similarly, we can construct a bit string of length eight ending with the two bits 00, in $2^6 = 64$ ways. This follows by the product rule because each of the first six bits can be chosen in two ways and the last

two bits can be chosen in only one way.

1	$2^7=128$ way		
0	$2^6=64$ ways	0	0
1	$2^5=32$ ways		0
0			

Some of the ways to construct a bit string of length eight starting with a 1 are the same as the ways to construct a bit string of length eight that ends with the two bits 00. There are $25 = 32$ ways to construct such a string. This follows by the product rule because the first bit can be chosen in only one way, each of the seconds through the sixth bits can be chosen in two ways, and the last two bits can be chosen in

one way. Consequently, the number of bit strings of length eight that begin with a 1 or end with a 00, which equals the number of ways to construct a bit string of length eight that begins with a 1 or that ends with 00, equals $128 + 64 - 32 = 160$. We present an example that illustrates how the formulation of the principle of inclusion-exclusion can be used to solve counting problems.



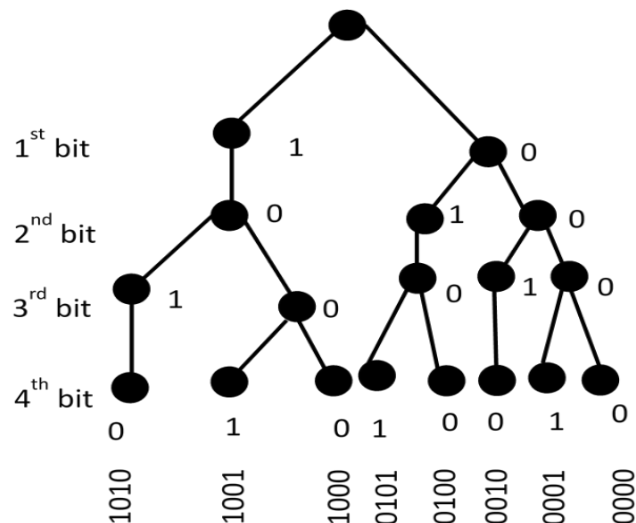
Example 19

A computer company receives 350 applications from computer graduates for a job planning a line of new Web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

Solution:

To find the number of these applicants who majored neither in computer science nor in business, we can subtract the number of students who majored either in computer science or in business (or both) from the total number of applicants. Let A_1 be the set of students who majored in computer science and A_2 the set of students who majored in business. Then $A_1 \cup A_2$ is the set of students who majored in computer science or business (or both), and $A_1 \cap A_2$ is the set of students who majored both in computer science and in business. By the subtraction rule, the number of students who majored either in computer science or in business (or both) equals

$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2| = 220 + 147 - 51 = 316$. We conclude that $350 - 316 = 34$ of the applicants majored neither in computer science nor in business. ▲ The subtraction rule, or the principle of inclusion-exclusion, can be generalized to find the number of ways to do one of n different tasks or, equivalently, to find the number of elements in the union of n sets, whenever n is a positive integer. We will study the inclusion-exclusion principle and some of its many applications in Chapter 8.



12.3 Tree Diagrams

FIGURE 2 Bit strings of length four without consecutive 1's

Counting problems can be solved using tree diagrams. A tree consists of a root, several branches leaving the root, and possible additional branches leaving the endpoints of other branches. To use trees in counting, we use a branch to represent each possible choice. We represent the possible outcomes by the leaves, which are the endpoints of branches not having other branches starting at them.

Note that when a tree diagram is used to solve a counting problem, the number of choices of which branch to follow to reach a leaf can vary.

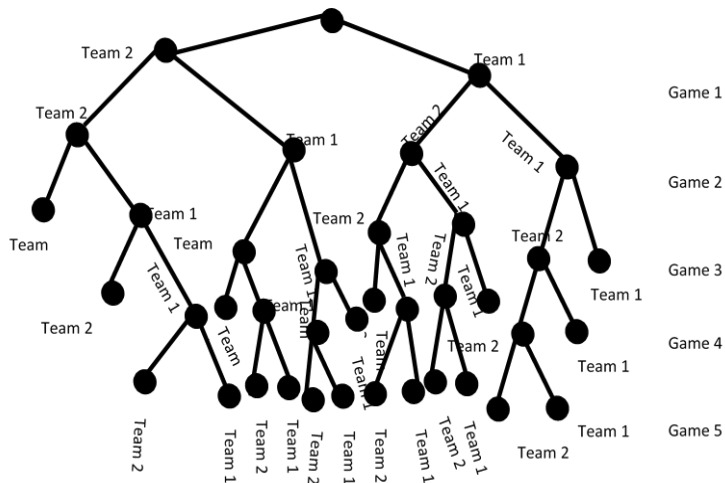


FIGURE 4. Best three games out of five playoff



Example 21

How many bit strings of length four do not have two consecutive 1s?

Solution:

The tree diagram in Figure 2 displays all bit strings of length four without two consecutive 1s. We see that there are eight-bit strings of length four without two consecutive 1s.



Example 22

A playoff between two teams consists of at most five games. The first team that wins three games wins the playoff. In how many different ways can the playoff occur?

Solution:

The tree diagram in Figure 3 displays all the ways the playoff can proceed, with the winner of each game shown. We see that there are 20 different ways for the playoff to occur.



Example 23

Suppose that “I Love New Jersey” T-shirts come in five different sizes: S, M, L, XL, and XXL.

Further suppose that each size comes in four colours, white, red, green, and black, except for XL, which comes only in red, green, and black, and XXL, which comes only in green and black. How many different shirts do a souvenir shop have to stock to have at least one of each available size and colour of the T-shirt?

Solution:

The tree diagram in Figure 4 displays all possible size and colour pairs. It follows that the souvenir shop owner needs to stock 17 different T-shirts.

12.4 Permutations and Combinations

Introduction Many counting problems can be solved by finding the number of ways to arrange a specified number of distinct elements of a set of a particular size, where the order of these elements matters. Many other counting problems can be solved by finding the number of ways to select a particular number of elements from a set of a particular size, where the order of the elements selected does not matter. For example, in how many ways can we select three students from a group of five students to stand in line for a picture? How many different committees of three students can be formed

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from a group of four students? In this section, we will develop methods to answer questions such as these. Permutations We begin by solving the first question posed in the introduction to this section, as well as related questions.



EXAMPLE 1

In how many ways can we select three students from a group of five students to stand in line for a picture? In how many ways can we arrange all five of these students in a line for a picture? Solution: First, note that the order in which we select the student's matters. There are five ways to select the first student to stand at the start of the line. Once this student has been selected, there are four ways to select the second student in the line. After the first and second students have been selected, there are three ways to select the third student in the line. By the product rule, there are $5 \cdot 4 \cdot 3 = 60$ ways to select three students from a group of five students to stand in line for a picture. To arrange all five students in a line for a picture, we select the first student in five ways, the second in four ways, the third in three ways, the fourth in two ways, and the fifth in one way. Consequently, there are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways to arrange all five students in a line for a picture.

The notion of permutation is used with several slightly different meanings, all related to the act of permuting (rearranging) objects or values. Informally, a permutation of a set of objects is an arrangement of those objects in a particular order. When doing permutation, the order of the items is important. For example, permutations of three items a, b, c are ab, ba, ac, ca, bc and cb . The number of permutations of n things taking r at a time is denoted by nP_r and the expression is

$${}^nP_r = \frac{n!}{(n-r)!}$$

Some other formulae on permutations are as follows:

The number of permutations of n different things taken r at a time is given by

$${}^nP_r = n^r$$

The number of permutations of n things taken all together, when x of the things is alike of one kind, y of the things is alike of one kind and z are rest of the things that are alike, is given by ${}^nP_r = \frac{n!}{x!y!z!}$

3. The number of ways of arranging n distinct objects along a round table is given by $(n-1)!$

4. The number of ways of arranging n persons along a round table so that no person has the same two neighbors is given by $\frac{1}{2}(n-1)!$



EXAMPLE 2

Let $S = \{1, 2, 3\}$. The ordered arrangement $3, 1, 2$ is a permutation of S . The ordered arrangement $3, 2$ is a 2-permutation of S . The number of r -permutations of a set with n elements is denoted by $P(n, r)$. We can find $P(n, r)$ using the product rule.



EXAMPLE 3

Let $S = \{a, b, c\}$. The 2-permutations of S are the ordered arrangements $a, b; a, c; b, a; b, c; c, a;$ and c, b . Consequently, there are six 2-permutations of this set with three elements. There are always six 2-permutations of a set with three elements. There are three ways to choose the first element of the arrangement. There are two ways to choose the second element of the arrangement because it must be different from the first element. Hence, by the product rule, we see that $P(3, 2) = 3 \cdot 2 = 6$. the first element. By the product rule, it follows that $P(3, 2) = 3 \cdot 2 = 6$.

We now use the product rule to find a formula for $P(n, r)$ whenever n and r are positive integers with $1 \leq r \leq n$.

THEOREM 1

If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are $P(n, r) = n(n-1)(n-2) \cdots (n-r+1)$ r -permutations of a set with n distinct elements. Proof: We will use the product rule to prove that this formula is correct. The first element of the permutation can be chosen in n ways because there are n elements in the set. There are $n-1$ ways to choose the second element of the permutation because there are $n-1$ elements left in the set after using the element picked for the first position. Similarly, there are $n-2$ ways to choose the third element, and so on, until there are exactly $n-(r-1) = n-r+1$ ways to choose the r^{th} element. Consequently, by the product rule, there are $n(n-1)(n-2) \cdots (n-r+1)$ r -permutations of the set. Note that $P(n, 0) = 1$ whenever n is a non-negative integer because there is exactly one way to order zero elements. That is, there is exactly one list with no elements in it, namely the empty list. We now state a useful corollary of Theorem 1. COROLLARY 1

If n and r are integers with $0 \leq r \leq n$, then $P(n, r) = \frac{n!}{(n-r)!}$. Proof: When n and r are integers with $1 \leq r \leq n$, by Theorem 1 we have $P(n, r) = n(n-1)(n-2) \cdots \frac{n!}{(n-r)!}$. Because $\frac{n!}{(n-0)!} = \frac{n!}{n!} = 1$ whenever n is a non-negative integer, we see that the formula $P(n, r) = \frac{n!}{(n-r)!}$ also holds when $r = 0$.

By Theorem 1 we know that if n is a positive integer, then $P(n, n) = n!$. We will illustrate this result with some examples.



EXAMPLE 4

How many ways are there to select a first-prize winner, a second-prize winner, and a third-prize winner from 100 different people who have entered a contest?

Solution:

Because it matters which person wins which prize, the number of ways to pick the three prize winners is the number of ordered selections of three elements from a set of 100 elements, that is, the number of 3-permutations of a set of 100 elements. Consequently, the answer is $P(100, 3) = 100 \cdot 99 \cdot 98 = 970,200$.



EXAMPLE 5

Suppose that there are eight runners in a race. The winner receives a gold medal, the second-place finisher receives a silver medal, and the third-place finisher receives a bronze medal. How many different ways are there to award these medals, if all possible outcomes of the race can occur and there are no ties?

Solution:

The number of different ways to award the medals is the number of 3-permutations of a set with eight elements. Hence, there are $P(8, 3) = 8 \cdot 7 \cdot 6 = 336$ possible ways to award the medals.



EXAMPLE 6

Suppose that a saleswoman has to visit eight different cities. She must begin her trip in a specified city, but she can visit the other seven cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities?

Solution:

The number of possible paths between the cities is the number of permutations of seven elements because the first city is determined, but the remaining seven can be ordered arbitrarily. Consequently, there are $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$ ways for the saleswoman to choose her tour. If, for instance, the saleswoman wishes to find the path between the cities with minimum distance, and she computes the total distance for each possible path, she must consider a total of 5040 paths!



EXAMPLE 7

How many permutations of the letters ABCDEFGH contain the string ABC?

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Solution:

Because the letters ABC must occur as a block, we can find the answer by finding the number of permutations of six objects, namely, the block ABC and the individual letters D, E, F, G, and H. Because these six objects can occur in any order, there are $6! = 720$ permutations of the letters ABCDEFGH in which ABC occurs as a block.

Combinations We now turn our attention to counting unordered selections of objects. We begin by solving a question posed in the introduction to this section of the chapter.



EXAMPLE 8

How many different committees of three students can be formed from a group of four students?

Solution:

To answer this question, we need only find the number of subsets with three elements from the set containing the four students. We see that there are four such subsets, one for each of the four students because choosing three students is the same as choosing one of the four students to leave out of the group. This means that there are four ways to choose the three students for the committee, where the order in which these students are chosen does not matter.

A combination is a way of selecting several things out of a larger group, where (unlike permutations) order does not matter. For example, combinations of three items a, b, c are ab, ac and bc .

Number of combinations of n things taking r at a time is denoted by nC_r and expression is

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Some other formulae on combinations are as follows:

The number of ways in which $a + b$ things can be divided into two groups containing a and b things, respectively, is

$${}^{a,b}C_b = \frac{(a+b)!}{a!b!}$$

The number of ways in which a selection can be made out of $a + b + c$ things of which a are alike of one kind, b is alike of another kind and remaining c are different is given by $(a+1)(b+1)2^c - 1$

The number of ways of arranging n distinct objects along a round table is given by

$$(n-1)!$$

The number of ways of arranging n persons along a round table so that no person has the same two neighbors is given by

$$\frac{1}{2}(n-1)!$$

Partitions

A combination is nothing but portioning a set into two subsets, one containing k objects and the other containing the remaining $n - k$ objects. Hence, in general the set $S = \{1, 2, \dots, n\}$ can be partitioned into r subsets.

Consider the following iterative situation leading to partition of S . We first select a subset of n_1 elements from S . Having chosen the first subset, we select second subset containing n_2 elements from the remaining $n - n_1$ elements and so on until no elements remain. This procedure yields a partition of S into r subsets, with the p^{th} subset containing exactly n_p elements.

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Now we wish to count the number of such partitions. We know that there are $\frac{n!}{n!(n-n_1)!}$ ways to form the first subset

Hence, we can formulate that there are $\frac{(n-n_1-n_2-\dots-n_{p-1})!}{n_p!(n-n_1-\dots-n_p)!}$ ways to form the p^{th} subset.

Using the counting principle, the total number of partitions is then given by $\frac{n!}{n_1!n_2!\dots n_r!}$

This expression is called a multinomial coefficient.

12.5 Introduction to Probability

The word probability or chance is very frequently used in day-to-day life. For example, we generally say, 'He may come today' or 'probably it may rain tomorrow' or 'most probably he will get through the examination'. All these phrases involve an element of uncertainty and probability is a concept which measures these uncertainties. The probability when defined in simplest way is the chance of occurring of a certain event when expressed quantitatively, i.e., probability is a quantitative measure of the certainty. The probability has its origin in the problems dealing with games of chance such as gambling, coin tossing, die throwing and playing cards. In all these cases the outcome of a trial is uncertain. These days probability is widely used in business and economics in the field of predictions for future. The following remarks may be important for learning this chapter on probability.

1. Die: A die is a small cube used in games of chance. On its 6 faces dots are marked as : : : :

Plural of die is dice. The outcome of throwing (or tossing) a die is the number of dots on its uppermost face. An ace on a die means 1 dot. 2. Cards: A pack (or deck) of playing cards has 52 cards, divided into 4 suits:

Spades (♠)

Clubs (♣)

Hearts (♥)

Diamonds (♦)

Each suit has 13 cards, nine cards numbered 2 to 10, an Ace (bDdk), a King (ckn'kkg), Queen (csxe) and a Jack or Knave (xqyke). Spades and Clubs are blackfaced cards while Hearts and Diamonds are red-faced cards. The Aces, Kings, Queens and Jacks are called face cards and other cards are called number cards. The Kings, Queens and Jacks are called court cards.

3. The number of combinations of n objects taken r at a time ($r \leq n$) is denoted by $C(n, r)$ or nC_r and is defined as

$$nC_r = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\dots \text{to } r \text{ factors}}{1.2.3\dots r.}$$

Illustration 1: ${}_{5}C_3 = \frac{5.4.3}{1.2.3} = 10$, ${}_{5}C_0 = 1$ and ${}_{5}C_5 = 1$

If $r > \frac{n}{2}$, then it is better to simplify nC_r as nC_{n-r}

Illustration 2: ${}_{52}C_{50} = {}_{52}C_{52-50} = {}_{52}C_2 = \frac{52.51}{2.1}$
 $= 26.51 = 1326$

When, $r > n$, $nC_r = 0$.

Some Important terms and concepts Random Experiment or Trial: The performance of an experiment is called a trial. An experiment is characterised by the property that its observations under a given set of circumstances do not always lead to the same observed outcome but rather to the different outcomes. If in an experiment all the possible outcomes are known in advance and none of the outcomes can be

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predicted with certainty, then such an experiment is called a random experiment. For example, tossing a coin or throwing a die are random experiments.

Event: The possible outcomes of a trial are called events. Events are generally denoted by capital letters A, B, C and so on.

Illustration 3:

When a coin is tossed the outcome of getting a head or a tail is an event.

When a die is thrown the outcome of getting 1 or 2 or 3 or 4 or 5 or 6 is an event. Sample Space: The set of all possible outcomes of an experiment is called a sample space. We generally denote it by S.

Illustration 4:

When a coin is tossed, $S = \{H, T\}$ where H = head, T = tail.

When a die is thrown, $S = \{1, 2, 3, 4, 5, 6\}$.

When 2 coins are tossed simultaneously, $S = \{HH, HT, TH, TT\}$ Equally Likely Events: Events are said to be equally likely if there is no reason to expect any one in preference to other. Thus, equally likely events mean outcome is as likely to occur as any other outcome.

Illustration 5:

In throwing a die, all the 6 faces (1, 2, 3, 4, 5, 6) are equally likely to occur. Simple and compound Events In the case of simple events we consider the probability of happening or non-happening of single events.

Illustration 6: We might be interested in finding out the probability of drawing an ace from a pack of cards. In the case of compound events we consider the joint occurrence of two or more events.

Illustration 7: If from a bag, containing 8 red and 5 green balls, two successive draws of 2 balls are made, we shall be finding out the probability of getting 2 red balls in the first draw and 2 green balls in the second draw. We are thus dealing with a compound event. Exhaustive Events: It is the total number of all possible outcomes of any trial.

Illustration 8:

When a coin is tossed, either head or tail may turn up and therefore, there are two exhaustive cases.

There are six exhaustive cases or events in throwing a die.

If two dice are thrown simultaneously, the possible outcomes are:

(1, 1) (2, 1) (3, 1) (4, 1) (5, 1) (6, 1)

(1, 2) (2, 2) (3, 2) (4, 2) (5, 2) (6, 2)

(1, 3) (2, 3) (3, 3) (4, 3) (5, 3) (6, 3)

(1, 4) (2, 4) (3, 4) (4, 4) (5, 4) (6, 4)

(1, 5) (2, 5) (3, 5) (4, 5) (5, 5) (6, 5)

(1, 6) (2, 6) (3, 6) (4, 6) (5, 6) (6, 6)

Thus, in this case, there are $36 (=6^2)$ ordered pairs. Hence, the number of exhaustive cases in the simultaneous throw of two dice is 36.

Three dice are thrown, the number of exhaustive cases is 6^3 , i.e., 216

algebra of Events

If A and B are two events associated with sample space S, then

$A \cup B$ is the event that either A or B or both occur.

$A \cap B$ is the event that A and B both occur simultaneously.

$\bar{A}$ is the event that A does not occur.

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(iv) $\overline{A \cap B}$ is an event of non-occurrence of both A and B, i.e., none of the events A and B occurs.

Illustration 9: In a single throw of a die, let A be the event of getting an even number and B be the event of getting a number greater than 2. Then,

$$A = \{1, 3, 5\}, B = \{3, 4, 5, 6\} \therefore A \cup B = \{1, 3, 4, 5, 6\}$$

$A \cup B$ is the event of getting an odd number or a number greater than 2. $A \cap B = \{3, 5\}$. $A \cap B$ is the event of getting an odd number greater than 2.

$\overline{A} = \{2, 4, 6\}$ [Those elements of S which are not in A.] $\overline{A}$ is the event of not getting an odd number, i.e., getting an even number.

$$\overline{B} = \{1, 2\}.$$

$\overline{B}$ is the event of not getting a number greater than 2, i.e., getting a number less than or equal to 2.

$$\overline{A \cap B} = \{2\}. \overline{A \cap B} \text{ is the event of neither getting an odd number nor a number greater than 2.}$$

Mutually Exclusive Events

In an experiment, if the occurrence of an event precludes or rules out the happening of all the other events in the same experiment.

Illustration 10:

When a coin is tossed either head or tail will appear. Head and tail cannot appear simultaneously. Therefore, occurrence of a head or a tail are two mutually exclusive events.

(ii) In throwing a die all the 6 faces numbered 1 to 6 are mutually exclusive since if any one of these faces comes, the possibility of others in the same trial, is ruled out.

Illustration 11:

(i) If the random experiment is 'a die is thrown' and A, B are the events, A: the number is less than 3; B: the number is more than 4, then $A = \{1, 2\}$, $B = \{5, 6\}$. $A \cap B = \phi$, thus A and B are mutually exclusive events.

(ii) If the random experiment is 'a card is drawn from a well-shuffled pack of cards' and A, B are the events A: the card is Black; B: the card is an ace. Since a black card can be an ace, $A \cap B \neq \phi$, thus A and B are not mutually exclusive events.

Mutually Exclusive and Exhaustive Events

Events $E_1, E_2, \dots, E_n$ are called mutually exclusive and exhaustive if $E_1 \cup E_2 \cup \dots \cup E_n = S$, i.e., $\bigcup_{i=1}^n E_i = S$ and $E_i \cap E_j = \phi$ for all $i \neq j$.

For example, in a single throw of a die, let A be the event of getting an even number and B be event of getting odd numbers, then

$$A = \{2, 4, 6\}, B = \{1, 3, 5\}$$

$$A \cap B = \phi, A \cup B = \{1, 2, 3, 4, 5, 6\} = S$$

$\therefore$ A and B are mutually exclusive and exhaustive events.

Illustration 12: Two dice are thrown and the sum of the numbers which come up on the dice noted. Let us consider the following events:

A: 'the sum is even'

B: 'the sum is a multiple of 3'

C: 'the sum is less than 4'

D: 'the sum is greater than 11'

Which pairs of these events are mutually exclusive? Solution: There are $6 \times 6 = 36$ elements in the sample space (Refer to Example 2).

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A is the event "the sum is even". It means we have to consider those ordered pairs (x, y) in which (x + y) is even. Thus, A = [(1, 1), (2, 2), (1, 3), (1, 5), (2, 4), (2, 6), (3, 1), (3, 3), (3, 5), (4, 2), (4, 4), (4, 6), (5, 1), (5, 3), (5, 5), (6, 2), (6, 4), (6, 6)].

Similarly, B = [(1, 2), (2, 1), (1, 5), (5, 1), (3, 3), (2, 4), (4, 2), (3, 6), (6, 3), (4, 5), (5, 4), (6, 6)]

C = [(1, 1), (2, 1), (1, 2)] D = [(6, 6)].

We find that $A \cap B = [(1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (6, 6)] \neq \phi$. Thus, A and B are not mutually exclusive.

Similarly, $A \cap C \neq \phi$, $A \cap D \neq \phi$, $B \cap C \neq \phi$, $B \cap D \neq \phi$, $C \cap D = \phi$. Thus, C and D are mutually exclusive.

Probability of an Event

The probability of an event is defined in the following two ways:

Mathematical (or a priori) definition

Statistical (or empirical) definition. Mathematical Definition of Probability: Probability of an event A, denoted as P(A), is defined as

$$P(A) = \frac{\text{Number of cases favourable to A}}{\text{Numbers of possible outcomes}}$$

Thus, if an event A can happen in m ways and fails (does not happen) in n ways and each of m + n ways is equally likely to occur then the probability of happening of the event A (also called success of A) is given by

$$P(A) = \frac{m}{m+n}$$

and that the probability of non-occurrence of the A (also called its failure) is given by

$$P(\text{not A}) \text{ or } P(\bar{A}) = \frac{n}{m+n}$$

If the probability of the happening of a certain event is denoted by p and that of not happening by q, then

$$p + q = \frac{m}{m+n} + \frac{n}{m+n} = 1$$

Here, p, q are non-negative and cannot exceed unity, i.e., $0 \leq p \leq 1$ and $0 \leq q \leq 1$. When p = 1, then the event is certain to occur. When p = 0, then the event is impossible. For example, the probability of throwing eight with a single die is zero.

Probability as defined above is sometimes called Priori Probability, i.e., it is determined before hand, that is, before the actual trials are made.

Illustration 13: A coin is tossed once. What are all possible outcomes? What is the probability of the coin coming up 'tails'?

Solution: The coin can come up either "heads" (H) or "tails" (T). Thus, the set S of all possible outcomes is $S = \{H, T\}$

$$\therefore P(T) = \frac{1}{2}$$

Illustration 14: What is the probability of getting an even number in a single throw of a die?

Solution: Clearly, a die can fall with any of its faces upper most. The number on each of the faces is, therefore, a possible outcome. Thus, there are total 6 outcomes. Since there are 3 even numbers on the die, namely, 2, 4 and 6,

$$P(\text{even number}) = \frac{3}{6} = \frac{1}{2}$$

Illustration 15: What is the probability of drawing a 'king' from a well-shuffled deck of 52 cards?

Solution: Well-shuffled ensures equally-likely outcomes. There are 4 kings in a deck. Thus,

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$$P(\text{a king}) = \frac{4}{52} = \frac{1}{13}$$

odds of an Event

Suppose, there are m outcomes favourable to a certain event and n outcomes unfavourable to the event in a sample space, then odds in favour of the event

$$= \frac{\text{Number of favourable outcomes}}{\text{Number of unfavourable outcomes}} = \frac{m}{n}$$

and odds against the event

$$= \frac{\text{Number of unfavourable outcomes}}{\text{Number of favourable outcomes}} = \frac{n}{m}$$

If odds in favour of an event A are $a:b$, then the probability of happening of event

$$\text{of happening of event } A = P(A) = \frac{a}{a+b} \text{ and probability of not happening of event } P(\bar{A}) = \frac{b}{a+b}$$

If odds against happening of an event A are $a:b$, then probability of happening of event $A = P(A) = \frac{b}{a+b}$

$$\text{and probability of not happening of event } A = P(\bar{A}) = \frac{a}{a+b}$$

Illustration 16: What are the odds in favour of getting a '3' in a throw of a die? What are the odds against getting a '3'?

Solution: There is only one outcome favourable to the event "getting" a 3, the other five outcomes, namely, 1, 2, 4, 5, 6 are unfavourable. Thus, Odds in favour of getting a '3'

$$= \frac{\text{Number of favourable outcomes}}{\text{Number of unfavourable outcomes}} = \frac{1}{5} \text{ or 1 to 5}$$

Odds against getting a '3'

$$\frac{\text{Number of favourable outcomes}}{\text{Number of unfavourable outcomes}} = \frac{1}{5} \text{ or 1 to 5}$$

Illustration 17: If the odds in favour of an event are 4 to 5, find the probability that it will occur.

Solution: The odds in favour of the event are $\frac{4}{5}$ thus,

$$\frac{P(A)}{1 - P(A)} = \frac{4}{5} \text{ i.e., } 4[1 - P(A)] = 5P(A),$$

$$\text{i.e., } P(A) = \frac{4}{9}$$

The probability that it will occur $= \frac{4}{9}$

fundamental theorems on Probability

Theorem 1: In a random experiment, if S is the sample space and E is an event, then

$$P(E) \geq 0$$

$$P(\phi) = 0$$

(iii) $P(S) = 1$. Remarks: It follows from above results that,

probability of occurrence of an event is always nonnegative;

probability of occurrence of an impossible event is 0;

probability of occurrence of a sure event is 1.

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Theorem 2: If E and F are mutually exclusive events, then,

$$P(E \cap F) = 0 \text{ and,}$$

$$(ii) P(E \cup F) = P(E) + P(F).$$

Theorem 3: If E and F are two mutually exclusive and exhaustive events, then $P(E) + P(F) = 1$.

Theorem 4: Let E be any event and be its complementary event, then $1 = P(\bar{E})$. Theorem 5: For any two events E and F, $P(E - F) = P(E) - P(E \cap F)$.

Theorem 6: (Addition Theorem). For any two events E and F,

$$P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

Notes

1. We may express the above results as $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$

2. If E and F are mutually exclusive, then $P(E \cap F) = 0$ and so $P(E \cup F) = P(E) + P(F)$.

Theorem 7: If E_1 and E_2 be two events such that $E_1 \subseteq E_2$, then prove that $P(E_1) \leq P(E_2)$. Theorem 8: If E is an event associated with a random experiment, then $0 \leq P(E) \leq 1$.

Theorem 9: For any three events E, F, G $P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E \cap F) - P(F \cap G) - P(E \cap G) + P(E \cap F \cap G)$

Illustration 18: A card is drawn at random from a wellshuffled pack of 52 cards. Find the probability of getting (i) a jack or a queen or a king, (ii) a two of heart or diamond.

Solution:

(i) In a pack of 52 cards, we have: 4 jacks, 4 queens and 4 kings.

Now, clearly a jack and a queen and a king are mutually exclusive events.

$$\text{Also, } P(\text{a jack}) = \frac{{}^4C_1}{{}^{52}C_1} = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{a queen}) = \frac{{}^4C_1}{{}^{52}C_1} = \frac{4}{52} = \frac{1}{13}$$

$$P(\text{a king}) = \frac{{}^4C_1}{{}^{52}C_1} = \frac{4}{52} = \frac{1}{13}$$

$\therefore$ By the addition theorem of Probability, $P(\text{a jack or a queen or a king}) = P(\text{a jack}) + P(\text{a queen}) + P(\text{a king})$

$$= \frac{1}{13} + \frac{1}{13} + \frac{1}{13} = \frac{3}{13}$$

(ii) $P(\text{two of heart or two of diamond}) = P(\text{two of heart}) + P(\text{two of diamond})$

$$= \frac{1}{52} + \frac{1}{52} = \frac{2}{52} = \frac{1}{26}$$

Illustration 19: Find the probability of getting a sum of 7 or 11 in a simultaneous throw of two dice.

Solution: When two dice are thrown we have observed that there are 36 possible outcomes. Now, we can have a sum of 7 as $1 + 6 = 7, 2 + 5 = 7, 3 + 4 = 7, 4 + 3 = 7, 5 + 2 = 7, 6 + 1 = 7$

Thus, the six favourable cases are (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)

$$\therefore P(\text{a sum of 7}) = \frac{6}{36} = \frac{1}{6}$$

Again, the favourable cases of getting a sum of 11 are (5, 6), (6, 5)

$$\therefore P(\text{a sum of 11}) = \frac{2}{36} = \frac{1}{18}$$

Since the events of getting 'a sum of 7' or 'a sum of 11' are mutually exclusive:

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$$\therefore P(\text{a sum of 7 or 11}) = P(\text{a sum of 7}) + P(\text{a sum of 11})$$

$$= \frac{1}{6} + \frac{1}{18} = \frac{4}{18} = \frac{2}{9}$$

Illustration 20: From a well-shuffled pack of 52 cards, a card is drawn at random, find the probability that it is either a heart or a queen.

Solution: A: Getting a heart card B: Getting a queen card

$$P(A) = \frac{13}{52}, P(B) = \frac{4}{52}, P(A \cap B) = \frac{1}{52}$$

$$\text{Required probability} = P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

Independent Events

Two event A and B are said to be independent if the occurrence (or non-occurrence) of one does not affect the probability of the occurrence (and hence non-occurrence) of the other.

Illustration 21: In the simultaneous throw of 2 coins, 'getting a head' on first coin and 'getting a tail on the second coin are independent events.

Illustration 22: When a card is drawn from a pack of well-shuffled cards and replaced before the second card is drawn, the result of second draw is independent of first draw. We now state, without proof, the theorem which gives the probabilities of simultaneous occurrence of the independent events.

Theorem 10: If A and B are two independent events, then $P(A \text{ and } B) = P(A) \cdot P(B)$

Illustration 23: Two dice are thrown. Find the probability of getting an odd number on the one die and a multiple of three on the other. Solution: Since the events of 'getting an odd number' on one die and the event of getting a multiple of three on the other are independent events,

$$P(A \text{ and } B) = P(A) \times P(B) \quad (1)$$

Now, $P(A) = P(\text{an odd number}) = \frac{3}{6} = \frac{1}{2}$ [There are 3 odd numbers 1, 3, 5] and $P(B) = P(\text{a multiple of 3}) = \frac{2}{6} = \frac{1}{3}$ [Multiples of 3 are 3 and 6]

$$\therefore \text{From (1), required probability} = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Illustration 24: Arun and Tarun appear for an interview for 2 vacancies. The probability of Arun's selection is $\frac{1}{3}$ and that of Tarun's selection is $\frac{1}{5}$. Find the probability that

only 1 of them will be selected,

none of them be selected. Solution: Let A: Arun is selected B: Tarun is selected.

Then, $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{5}$ Clearly

'A' and 'not B' are independent also 'not A' and 'not B' are independent, 'B' and 'not A' are independent. (i) $P(\text{only 1 of them will be selected})$

$$= P(A \text{ and not } B \text{ or } B \text{ and not } A) = P(A) P(\text{not } B) + P(B) P(\text{not } A)$$

$$= \frac{1}{3} \left(1 - \frac{1}{5}\right) + \frac{1}{5} \left(1 - \frac{1}{3}\right)$$

$$= \frac{1}{3} \times \frac{4}{5} + \frac{1}{5} \times \frac{2}{3} = \frac{4}{15} + \frac{2}{15}$$

$$= \frac{6}{15} = \frac{2}{5}$$

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P (onle 1 of them be selected)

$$= P(\text{not A and not B}) = P(\text{not A}) \times P(\text{not B})$$

$$= \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{5}\right)$$

$$= \frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

12.6 Bayes' Rule

$$P(A | B) = P(A \cap B) / P(B)$$

- Allows us to express $P(A | B)$ in terms of $P(B | A)$, which may be easier to work with.



Example: Machine 1 makes 30% of all parts in a factory, and Machine 2 makes the rest. 2% of all parts made by Machine 1 are defective, and 3% of all parts made by Machine 2 are defective. Suppose we find a part that is defective. What is the probability that it came from Machine 1?



Example 2

In Orange County, 51% of the adults are males. (It doesn't take too much advanced mathematics to deduce that the other 49% are females.) One adult is randomly selected for a survey involving credit card usage.

- Find the prior probability that the selected person is a male.
- It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration). Use this additional information to find the probability that the selected subject is a male.

Solution

Let's use the following notation:

M = male

$\bar{M}$ = female (or not male)

C = cigar smoker

$\bar{C}$ = not a cigar smoker.

- Before using the information given in part b, we know only that 51% of the adults in Orange County are males, so the probability of randomly selecting an adult and getting a male is given by $P(M) = 0.51$.

- Based on the additional given information, we have the following:

$P(\bar{M}) = 0.51$ because 51% of the adults are males

$P(M) = 0.49$ because 49% of the adults are females (not males)

$P(C | M) = 0.095$ because 9.5% of the males smoke cigars (That is, the probability of getting someone who smokes cigars, given that the person is a male, is 0.095.)

Let's now apply Bayes' theorem by using the preceding formula with M in place of A, and C in place of B. We get the following result:

$$\begin{aligned}P(M|C) &= \frac{P(M) \cdot P(C|M)}{[P(M) \cdot P(C|M)] + [P(\overline{M}) \cdot P(C|\overline{M})]} \\&= \frac{0.51 \cdot 0.095}{[0.51 \cdot 0.095] + [0.49 \cdot 0.017]} \\&= 0.85329341 \\&= 0.853 \text{ (rounded)}\end{aligned}$$

Before we knew that the survey subject smoked a cigar, there is a 0.51 probability that the survey subject is male (because 51% of the adults in Orange County are males). However, after learning that the subject smoked a cigar, we revised the probability to 0.853. There is a 0.853 probability that the cigar-smoking respondent is a male. This makes sense, because the likelihood of a male increases dramatically with the additional information that the subject smokes cigars (because so many more males smoke cigars than females).

Summary

The key concepts learned from this unit are: -

- We have learned what are the basics of counting.
- We have learned how to use the product rule.
- We have learned the use of the product rule in the basics of counting.
- We have learned the use of the Sum rule in the basics of counting
- We have learned the use of the Subtraction rule in the basics of counting.
- We have learned the use of the product rule, the Sum Rule and the Subtraction rule in the basics of counting.
- We have learned the use of a tree diagram in the basics of counting.
- We have learned the Division rule in the basics of counting.
- We have learned how to find the total number of functions in the basics of counting
- We have learned how to count the different possible ways of combination.
- We have learned some counting related to the deck of cards with different examples
- We have learned what a pigeonhole principle is.
- We have learned with different examples how to count the different possible ways of combination.

Keywords

- basics of counting.
- Product rule
- Sum rule
- Subtraction rule
- tree diagram
- Division rule
- Combination.
- Deck of cards
- Pigeonhole principle.

Self Assessment

1. There are four bus lines between A and B and three bus lines between B and C , In how many ways can a man travel by bus from A to C by way of B ?
A. 12
B. 13
C. 14
D. 15
2. Suppose repetitions are not permitted. How many three- digit numbers can be formed from the six digits 2,3,5,6,7 and 9
A. 120

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- B. 140
C. 150
D. 200
3. Find the number of ways that a party of seven persons can arrange themselves: in a row of seven chairs.
- A. 15!
B. 17!
C. 20!
D. 22!
4. The number of distinct permutations that can be formed from all the letters of each word :
RADAR is
- A. 30
B. 40
C. 50
D. 60
5. In how many ways can four mathematics books, three history books ,three chemistry books. And
two sociology books be arranged on a shelf so that all books of the same subject are together?
- A. 350
B. 450
C. 472
D. 879
6. If $p(n,2) = 72$, then $n =$
- A. 9
B. 8
C. 7
D. 5
7. In how many ways are find examinations be scheduled in a week so that no two examinations
are scheduled on the same day considering Sunday as a holidays?
- A. 15
B. 12
C. 10
D. 6
8. In a certain programming language ,variables should be of length three and should be made up
of two letters followed by a digit or of length two made up of a letter followed by a digit. How
many possible variables ?

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- A. 250
- B. 260
- C. 720
- D. 200

9. Six boys and 6 girls are to be seated in a row , how many ways can they be seated if All boys are to be seated together and all girls are to be seated together.

- A. $6! \times 6!$
- B. $6! \times 2$
- C. $6! \times 6! \times 2$
- D. $6! \times 6! \times 2 \times 2$

10. How many ways can be letter in word MISSISSIPPI can be arranged ?

- A. $\frac{11!}{4!4!11!}$
- B. $\frac{4!2!}{10!}$
- C. $\frac{4!4!2!}{11!}$
- D. $\frac{4!4!2!}{11!}$

11. In how many ways can letters a, b, c, d, m and n be arranged if M and n always appear together

- A. $10! \times 3$
- B. $5! \times 2$
- C. $5! \times 5$
- D. $10! \times 4$

12. If repetitions are not permitted ,how many four digit number can be formed from digit 1,2,3,7,8, and 5. Total number of less than 5000.

- A. 100
- B. 180
- C. 280
- D. 380

13. A word that reads the same when read in forward or backward is called as palindrome. How many seven -letters palindromes can be formed from english alphabets?

- A. 26^2
- B. 25^4
- C. 26^4
- D. 26^5

14. Suppose repetitions are not permitted. How many of these numbers are less than 400?

- A. 30

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- B. 40
- C. 50
- D. 60

15. Find the number of ways that a party of seven persons can arrange themselves: around a circular table.

- A. $(n)!$
- B. $(n-1)!$
- C. $(n-2)!$
- D. $(n-3)!$

16. A woman has 11 close friends. In how many ways can she invite five of them to dinner?

- A. 456;
- B. 462;
- C. 450;
- D. 451;

17. A man has 11 close friends. In how many ways if two of the friends are married and will not attend separately?

- A. 210;
- B. 222;
- C. 223;
- D. 234;

18. A person has 11 close friends. In how many ways if two of them are not on speaking terms and will not attend together?

- A. 456
- B. 786
- C. 897
- D. 252

19. A woman has 11 close friends of whom six are also women. In how many ways can she invite three or more to a party?

- A. $2^{11} - 1 - \binom{11}{2} - \binom{11}{2} = 1981$
- B. $2^9 - 1 - \binom{17}{2} - \binom{17}{2} = 1980$

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C. $2^7 - 1 - \binom{15}{2} - \binom{15}{2} = 1982$

D. $2^5 - 1 - \binom{10}{2} - \binom{10}{2} = 1983$

20. A woman has 11 close friends of whom six are also women. In how many ways can she invite three or more of them if she wants the same number of men as women (including herself)?

- A. 325
- B. 360
- C. 400
- D. 480

21. A student is to answer 10 out of 13 questions on an exam. How many choices has he?

- A. 567
- B. 456
- C. 786
- D. 286

22. A student is to answer 10 out of 13 questions on an exam. How many if he must answer the first two questions?

- A. 897
- B. 890
- C. 165
- D. 987

23. A student is to answer 10 out of 13 questions on an exam. How many if he must answer the first or second question but not both?

- A. 675
- B. 876
- C. 110
- D. 654

24. A student is to answer 10 out of 13 questions on an exam. How many if he must answer exactly three out of the first five question?

- A. 876
- B. 80

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- C. 87
D. 90

25. A student is to answer 10 out of 13 questions on an exam. How many if he must answer at least three of the five questions?

- A. 897
B. 276
C. 234
D. 543

26. The minimum number of students needed to guarantee that five of them belong to the same class (freshman, sophomore, junior, senior)

- A. Here the $n=3$ classes are the pigeonholes and $k+1=7$ so $k=8$. Thus, among any $kn+1=14$ student (pigeons), five of them belong to the same class.
B. Here the $n=4$ classes are the pigeonholes and $k+1=5$ so $k=4$. Thus, among any $kn+1=17$ student (pigeons), five of them belong to the same class.
C. Here the $n=6$ classes are the pigeonholes and $k+1=50$ so $k=41$. Thus, among any $kn+1=107$ student (pigeons), five of them belong to the same class.
D. Here the $n=24$ classes are the pigeonholes and $k+1=15$ so $k=4$. Thus, among any $kn+1=17$ student (pigeons), five of them belong to the same class.

27. Let L be a list (not necessarily in alphabetical order) of the 26 letters in the english alphabet (which consists of 5 vowels, A, E, I, O, U and 21 consonants) then L has a sublist consisting

- A. of four or more consecutive consonants.
B. of five or more consecutive consonants
C. of six or more consecutive consonants
D. of eight or more consecutive consonants

28. Let L be a list (not necessarily in alphabetical order) of the 26 letters in the english alphabet (which consists of 5 vowels, A, E, I, O, U and 21 consonants). Assuming L begins with a vowel, say A, then L has a sublist consisting

- A. of four or more consecutive consonants.
B. of five or more consecutive consonants
C. of six or more consecutive consonants

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D. of eight or more consecutive consonants

29. The minimum number n of integers to be selected from $S = \{1, 2, \dots, 9\}$ so that the sum of two of the n integers is even is.....

- A. 3
- B. 4
- C. 5
- D. 6

30. The minimum number n of integers to be selected from $S = \{1, 2, \dots, 9\}$ so that the difference of two of the n integers is 5 is

- A. 5
- B. 6
- C. 7
- D. 8

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. A | 2. A | 3. B | 4. A | 5. C |
| 6. A | 7. D | 8. B | 9. C | 10. D |
| 11. B | 12. B | 13. C | 14. B | 15. B |
| 16. B | 17. A | 18. D | 19. A | 20. A |
| 21. D | 22. C | 23. C | 24. B | 25. B |
| 26. B | 27. A | 28. B | 29. A | 30. B |

Review Questions

1. How many words can be formed out of the letters of the word "UNITED" taking all the letters at a time considering no letter is to be repeated?

Solution: The word "UNITED" consists of 6 different letters. Hence the required number of permutations

$$= {}^6P_6 = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

Therefore, the required number of ways is

$$({}^8C_2 \times {}^4C_2) + ({}^8C_1 \times {}^4C_3) + ({}^4C_4) = \left(\frac{8 \times 7}{2} \times \frac{4 \times 3!}{4}\right) + (8 \times 4) + (1)$$

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$$=168+33+1=201$$

2. From a group of 8 men and 4 women, 4 persons are to be selected to form a committee so that at least 2 women are there on the committee. In how many ways can it be done?

Solution: We may have (2 men and 2 women) or (1 man and 3 women) or (0 man and 4 women).

3. How many words can be formed from the letters of the word "COMMUTE" taking all the letters at a time considering no letter is to be repeated and the first and last letter of the words are "M"?

Solution: The word "COMMUTE" has two M's. The first and the last letter should be M. The remaining letters are C, O, U, T, E. So, the number of permutations is

$${}^5P_5 = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

4. In how many ways 6 prizes can be given to 10 students when a student may receive at most two prizes?

Solution: The first prize can be given in 10 ways as any student can receive it. Also, since a student may receive at most two prizes, the second prize can be given in 10 ways. Now, the third prize can be given in 9 ways since a student cannot get more than two prizes. Similarly, the fourth prize can be given in 9 ways. The fifth and sixth prize can be given in 8 ways each. Hence, the total number of ways in which the prizes can be given to 10 students when no student can get more than two prizes = $10 \times 10 \times 9 \times 9 \times 8 \times 8 = 518400$

5. How many natural numbers less than 1000 can be formed with the digits 1, 2, 3, 4 and 5, considering repetition of digits is not allowed?

Solution: The numbers can be of 1, 2 or 3 digits. The numbers can be formed using the digits 1, 2, 3, 4, 5. So the total number of natural numbers less than 1000 using 1, 2, 3, 4, 5 is

$$=5 + 5 \times 4 + 5 \times 4 \times 3 = 5 + 20 + 60 = 85$$

6. In how many ways can 5 beads of different colors form a necklace?

Solution: Here, we are to form circular permutations of 5 things taken all at a time. This can be done in $(5-1)!$ ways. But in case of a necklace, clockwise and anti-clockwise, permutations are same. Hence, the required number of ways is

$$\frac{1}{2}(5-1)! = \frac{1}{2}4! = \frac{1}{2} \times 24 = 12$$

7. Mary has 4 bananas, 6 oranges and 5 apples. How many selections of fruits can be made by selecting at least one of them?

Solution: Five apples can be selected in 6 different ways, that is, we can choose 1, 2, 3, 4, 5 or none of the apple. Similarly, 4 bananas can be selected in 5 different ways and 6 oranges can be selected in 7 different ways. Therefore, Number of ways = $6 \times 5 \times 7 = 210$

This also includes the case in which none of the fruits is selected. Rejecting this case, we get required number of ways = $210 - 1 = 209$.

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8. In how many ways can the word "APOCALYPSE" be arranged such that all the vowels come together?

Solution: The word "APOCALYPSE" has 10 letters. Now, all the vowels are supposed to come together, we consider all the vowels "A, O, A, E" as one. Hence, number of ways this can be arranged in

$(7! / 2!)$ (since 4 vowels are treated as 1, the number of arrangements become $7!$ and because "P" is repeated twice). Also, the vowels can be arranged in $4!$ number of ways within themselves but the vowel "A" is repeated twice. Hence, the total number of arrangements of the word "APOCALYPSE" such that all the vowels come together

$$\frac{7!}{2!} \times \frac{4!}{2!} = 7! \times 3! = 5040 \times 6 = 30240$$

9. If ${}^{n-1}P_3 : {}^nP_4 = 1:6$, then what is the value of n ?

Solution: We have

$${}^{n-1}P_3 : {}^nP_4 = 1:6$$

$$\Rightarrow \frac{(n-1)!}{(n-1-3)!} \times \frac{(n-4)!}{(n1)!} = \frac{1}{6}$$

$$\Rightarrow \frac{(n-1)!}{(n-4)!} \times \frac{(n-4)!}{(n)!} = \frac{1}{6}$$

$$\Rightarrow \frac{(n-1)!}{n!} = \frac{1}{6}$$

$$\Rightarrow \frac{(n-1)!}{n.(n-1)!} = \frac{1}{6}$$

$$\Rightarrow n=6$$

10. How many triangles can be formed by joining the Vertices of an octagon?

Solution: Total number of vertices in an octagon = 8. We know that a triangle is formed by joining three vertices. Hence, this can be done in 8C_3 number of ways. Thus, the total number of triangles that can be formed is

$${}^8C_3 = \frac{8!}{3! \times 5!} = \frac{8 \times 7 \times 6}{3 \times 2} = 56.$$

11. Record plant location (either 1, 2, 3) of next maintenance call for spinning machine repair:

Sample space =

12. Toss 2 fair coins:

Sample space =

13. An engineering design firm is up for a Nissan contract and a Ford contract:

Sample space =

Unit 12: Permutation and Combination, Probability

14. Plant 1 has 22 machines, plant 2 has 60 machines, plant 3 has 18 machines. Probability of next repair being from plant 3, denoted:

$$P(3) =$$

15. Event A = 'next repair from odd-numbered plant'

$$P(A) =$$

16. Event B = 'get at least one contract'

$$P(B) =$$

17. A manufacturer (Acme) of automobile lamps categorizes the lamps produced as "Good", "Satisfactory", or "Unsatisfactory" for both *intensity* and *useful life*. For 200 lamps, consider the table:

Useful Life

<i>Intensity</i>	Good	Satis	Unsatis	Total
Good	100	25	5	
Satis	35	10	5	
Unsatis	10	8	2	
Total				200

Unit 12: Permutation and Combination, Probability

Select a lamp at random. What is the probability that it will be rated “Good” in *intensity*?

A =

P(A) =

What is the probability that it will be rated “Good” in *useful life*?

B =

P(B) =

What is the probability that it will be rated “Good” in *intensity* and *useful life*?

What is the probability that it will be rated “Good” in *useful life* or *intensity*?

18. Suppose we toss 10 coins. Define event A = {obtain at least 2 “heads”}.

What is P(A)?

18. What is the probability that the lamp is Good in *intensity* given that the lamp is Good in *useful life*?

19. What is the probability that the lamp is Good in *intensity* given that the lamp is Unsatisfactory in *useful life*?

20. An auto-parts store purchases 30% of its lamps from the Acme manufacturer. Estimate the probability that a randomly chosen lamp in the store is from Acme and is “Unsatisfactory” in *intensity*.

21. When a computer goes down, there is a 75% chance that it is due to an overload and a 15% chance that it is due to a software problem. There is a 15% chance that *neither* an overload nor a software problem is the cause. For a random computer malfunction, what is the probability that both an overload and a software problem are liable?

22. 80% of accidents at a foundry involve human error and 40% involve equipment malfunction. 35% involve both problems. If an accident involves an equipment malfunction, what is the probability that there was also human error?

23. Acme example: A = {Good in *intensity*}, B = {Good in *useful life*}.

Are A and B independent?

24. Roll 1 die. Let A = {even number} and B = {less than 4}. Are A and B independent?

25. If two events A and B are mutually exclusive (and $P(A) > 0$ and $P(B) > 0$), then they cannot be independent. Why?

26. Four electrical components are connected in series. The reliability of each component is 0.90. If the components are independent, what is the probability that the circuit works when the switch is thrown?

27. Baye’s Theorem

	Positive Test Result (Pregnancy is indicated)	Negative Test Result (Pregnancy is not indicated)
Subject is Pregnant	80	5
Subject is Not Pregnant	3	11

1. a. If one of the 99 test subjects is randomly selected, what is the probability of getting a subject who is pregnant?

b. A test subject is randomly selected and is given a pregnancy test. What is the probability of getting a subject who is pregnant, given that the test result is positive?

Ans: a. 85/99 or 0.859

b. 80/83 or 0.964



Further Readings

1. Quantitative Aptitude For Competitive Examinations By Dr. R S Aggarwal, S Chand Publishing
2. A Modern Approach To Verbal & Non-Verbal Reasoning By Dr. R S Aggarwal, S Chand Publishing
3. Magical Book On Quicker Maths By M Tyra, Banking Service Chronicle
4. Analytical Reasoning By M.K. Pandey, Banking Service Chronicle